RETAIL SHOPPING INSIGHTS

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1. PROJECT OVERVIEW

This project explores customer shopping behaviour in a retail store to identify purchasing patterns, gauge sales performance, and better understand the factors and characteristics that contribute to customer spending. The analysis focuses on revenue generation, product performance, discount usage, shipping preferences, customer loyalty, and newsletter subscriptions. The objective is to demonstrate how data analytics can transform raw retail data into actionable business insights that can support informed decision-making.

The dataset was sourced from an existing GitHub repository and cleaned and modified using Python in Jupyter Notebook. The prepared dataset was then imported into MySQL, where SQL queries were used to investigate key business questions and analyse customer purchasing patterns. Finally, MySQL was connected to Power BI to create an interactive dashboard presenting selected findings through visualisations and key performance indicators (KPIs).

2. DATASET SUMMARY

Rows: 3900

Columns: 18

Key Features:

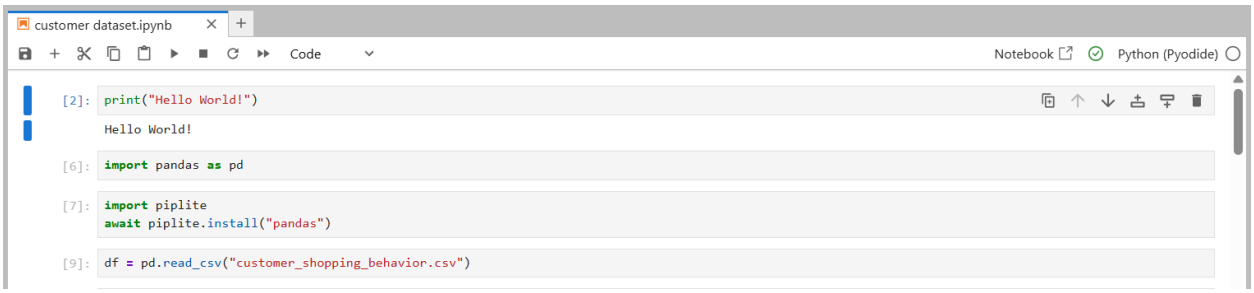
- Customer demographics (age, gender, location, subscription status)
- Purchase details (item purchased, category, purchase amount, season, size, colour)
- Shopping behaviour (discount applied, promo code used, previous purchases, frequency of purchases, review rating, shipping type)

Missing data: 37 values in the Review Rating column

3. EXPLORATORY DATA ANALYSIS USING PYTHON

Started off with data cleaning and modification in Python.

- **Data loading:** imported the dataset using Pandas



- **Initial data exploration:** Used `df.info()` to check data structure and `df.describe(include='all')` for summary statistics.

customer dataset.ipynb

Code | Notebook | Python (Pyodide)

Shipping

[12]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
```

customer dataset.ipynb

Code | Notebook | Python (Pyodide)

```
memory usage: 350.5 KB
```

[14]:
```
df.describe(include='all')
```

[14]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing data handling:** Checked for any null values. `Review Rating` had 37 missing values so I credited the missing values in the column by using the median of each product category.

customer dataset.ipynb

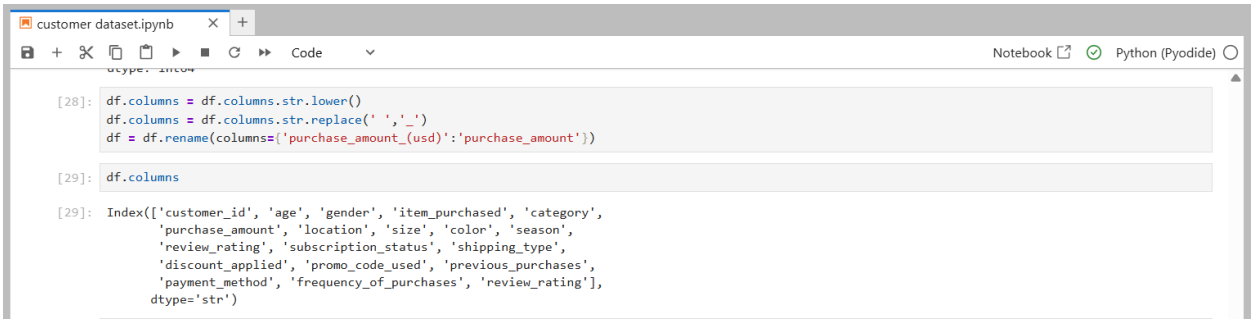
Code | Notebook | Python (Pyodide)

[15]:
```
df.isnull().sum()
```

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

[22]:
```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

- **Column standardisation:** Renamed columns and introduced snake case to improve readability and overall documentation.



- **Feature engineering:**
  Created `age_group` column by binning customer ages.



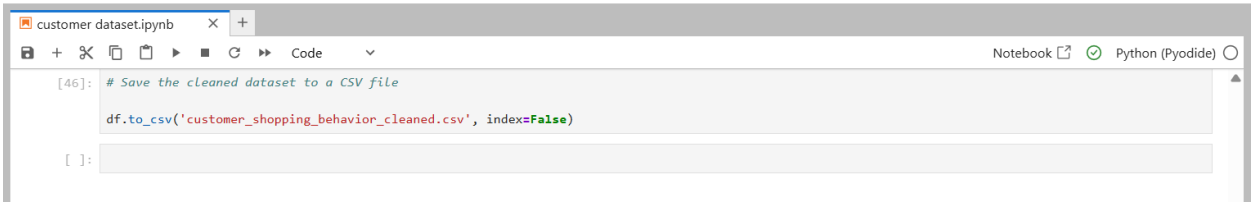
Created `purchase_frequency_days` column from purchase data.



- **Data consistency check:** verified whether `discount_applied` and `promo_code_used` were similar thus making the other column redundant. Dropped `promo_code_used`.



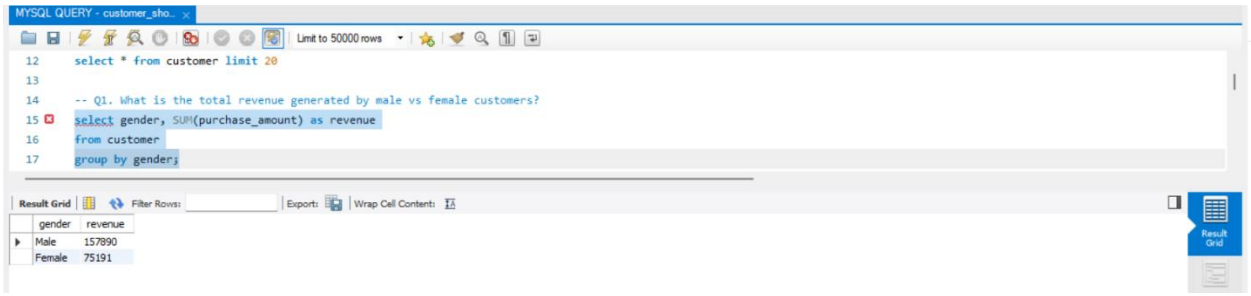
- **Save:** saved the cleaned and modified dataset into a csv file.



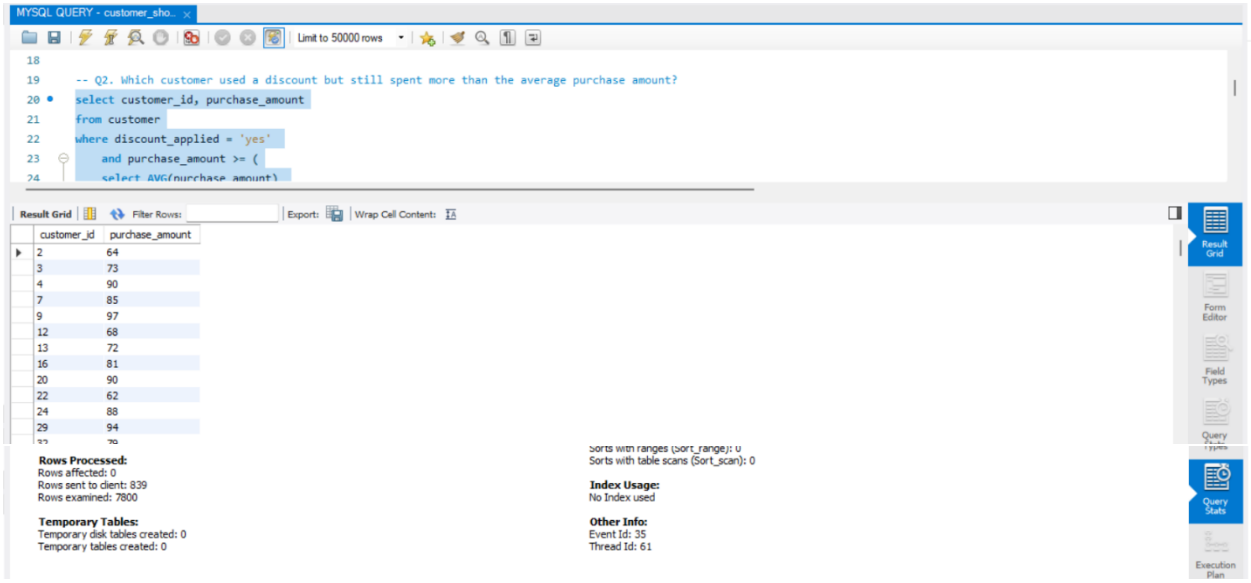
4. DATA ANALYSIS USING SQL

Created a new schema and uploaded the csv file onto MySQL and conducted structured data analysis to answer key business questions.

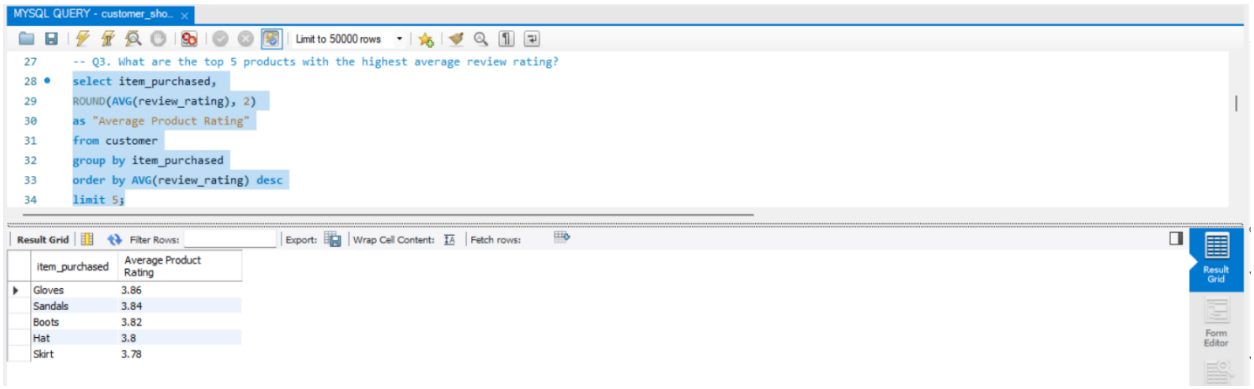
- **Revenue by Gender:** compared the total revenue generated by male customers vs female customers.



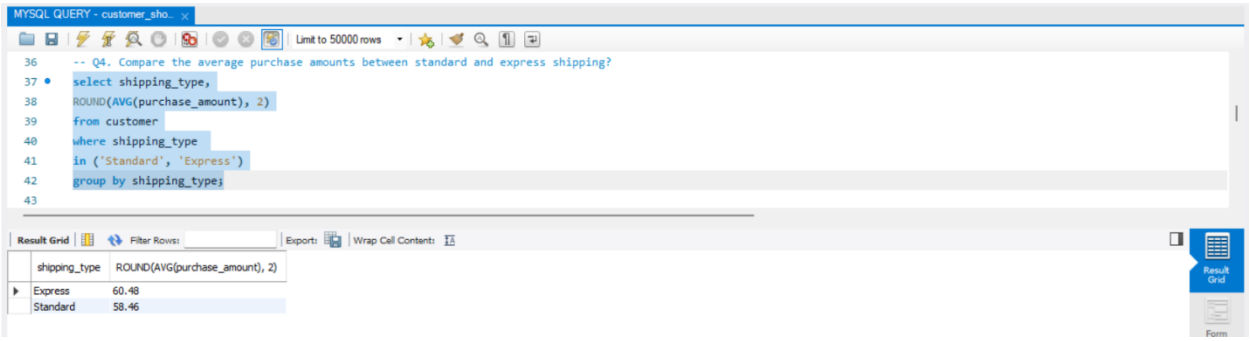
- **High-spending discount users:** determine which customers used discounts and still spent above the average purchase amount.



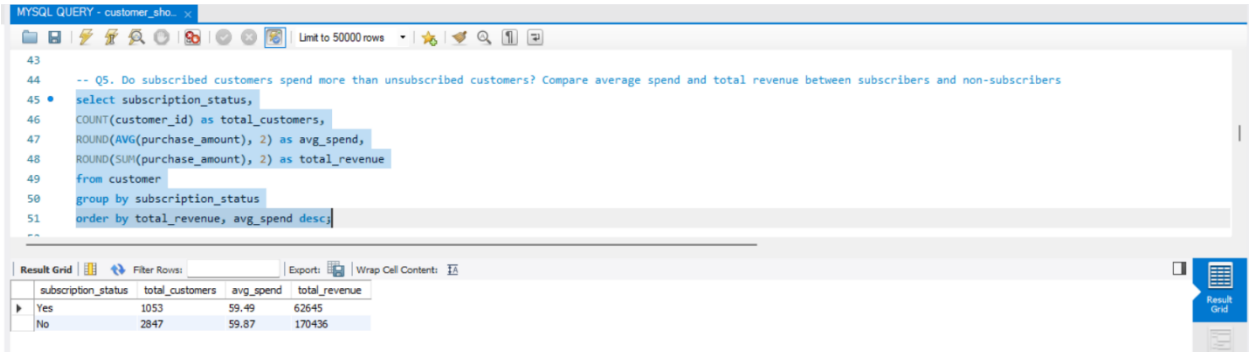
- **Top five product rating:** determine which products have the highest average review rating.



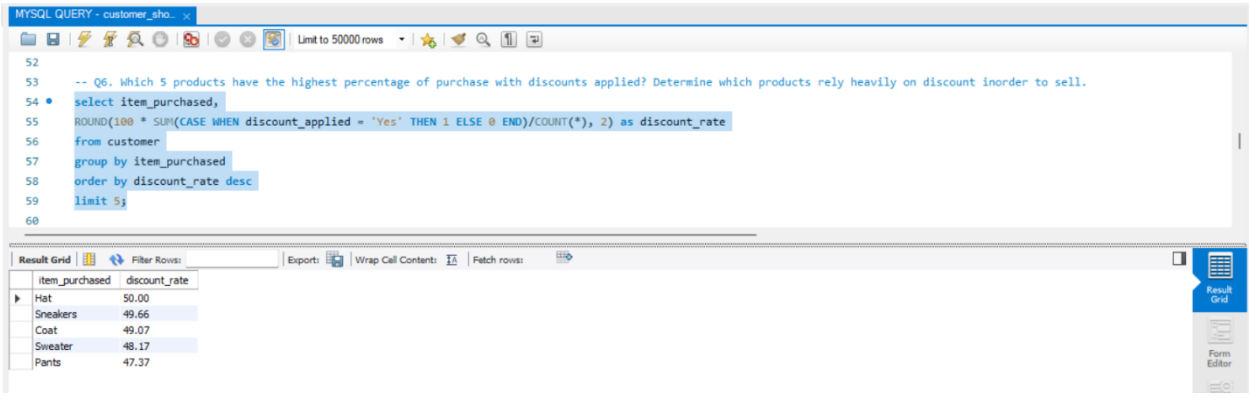
- **Shipping type comparison:** compared the average purchase amounts between the different shipping types.



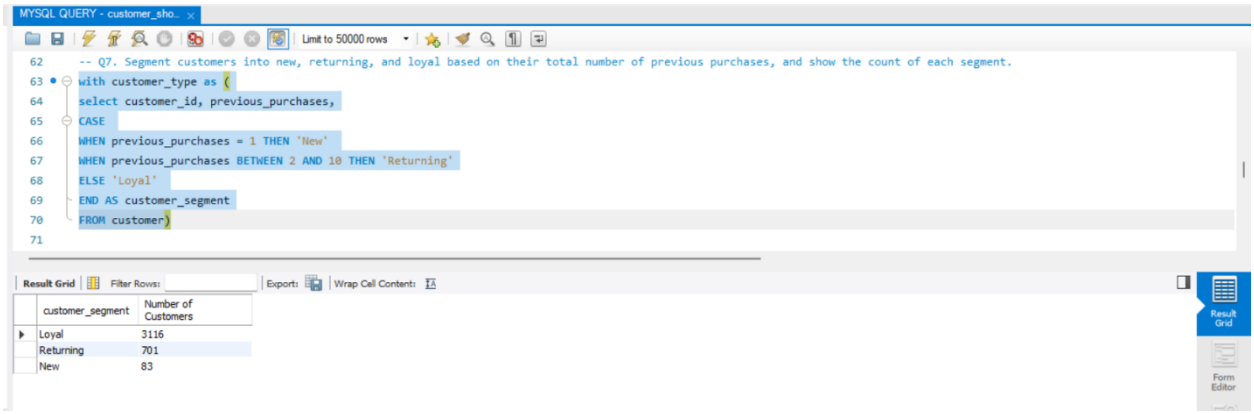
- **Newsletter subscriber's vs non-subscribers:** compared the average spend and total revenue across subscription status.



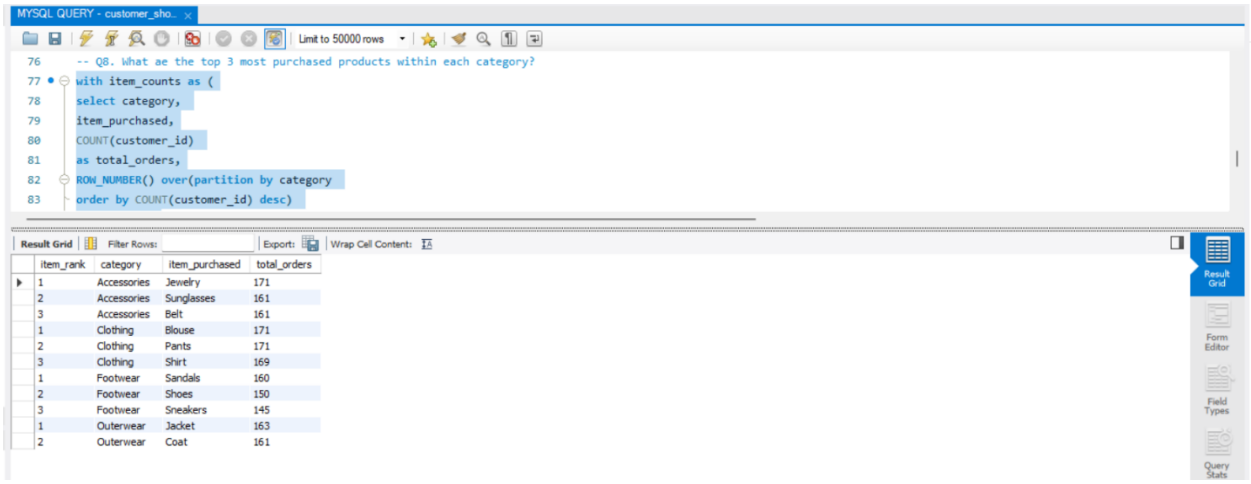
- **Discount dependent products:** determine which five products have the highest percentage of discounted purchases.



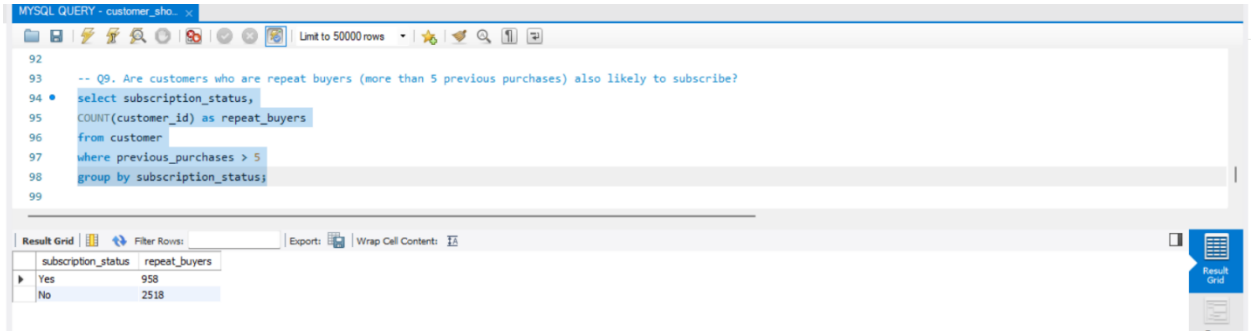
- **Customer segregation:** classify customers into three categories (New, Returning, and Loyal) based on their purchase history.



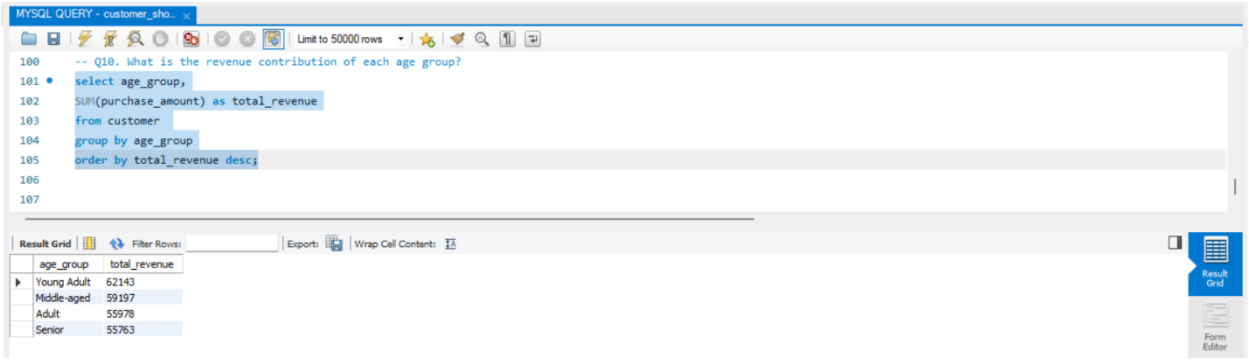
- **Top three products per category:** identify the three most purchased products in each category.



- **Repeat buyers & Subscriptions:** determine the likelihood of customers with purchases higher than five whether they are likely to subscribe to the newsletter or not.

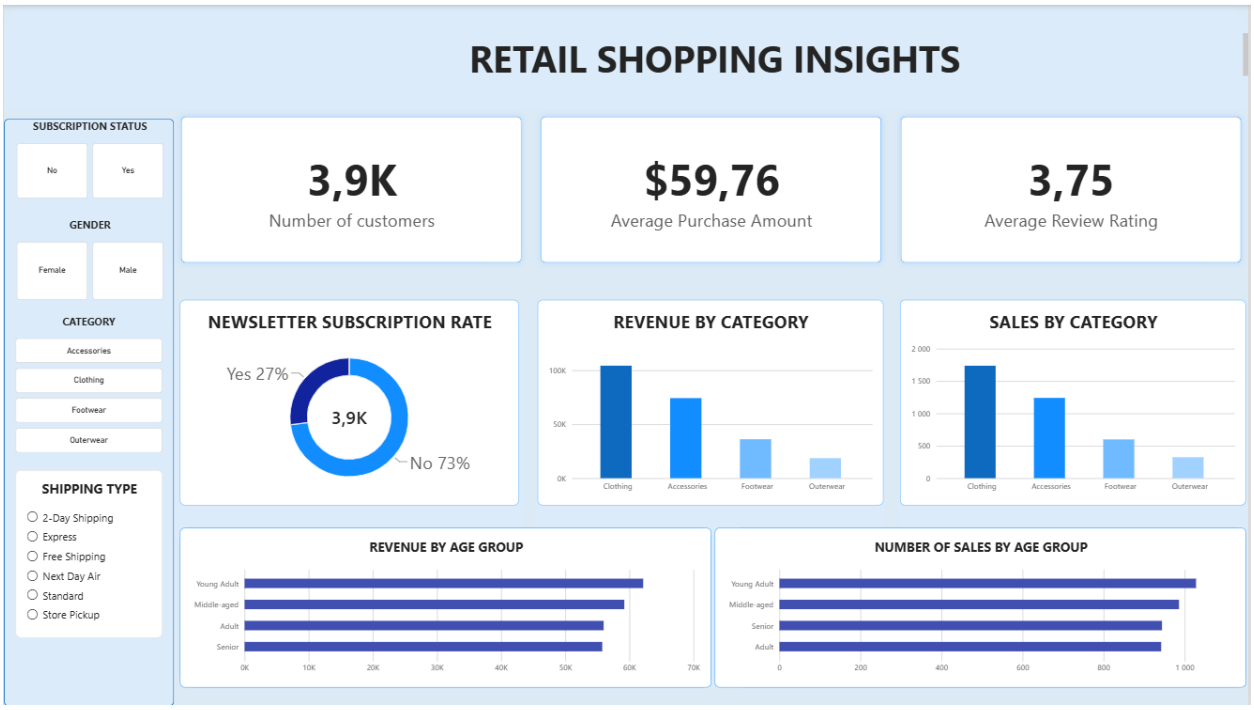


- **Revenue by age group**: calculated the total revenue contribution of each age group.



5. CREATE VISUALISATION DASHBOARD USING POWER BI

An interactive dashboard was developed in Power BI to present the key findings of the retail customer shopping behavioural analysis. The dashboard consolidates important metrics, including total customers, average purchase amount, customer ratings, newsletter subscription rates, revenue and sales by product category, and purchasing patterns across customer segments. Interactive filters allow users to explore the data by subscription status, gender, product category, and shipping type, making it easier to identify trends and compare customer behaviour.



6. BUSINESS RECOMMENDATIONS

Based on the SQL analysis, the following business recommendations are presented:

- **Review marketing strategies by gender:** Male customers generated $157890 in revenue compared with $75191 from female customers. Investigate the difference in customer numbers and purchasing patterns to identify opportunities for growth across both segments.

- **Review express shipping performance:** Customers using express shipping had a higher average purchase amount (60.48) compared to those using standard shipping (58.48). Identify whether premium shipping options can support customer satisfaction and revenue without greatly increasing fulfilment costs.

- **Improve newsletter subscription incentives:** Non-subscribed customers recorded a higher average purchase amount (58.87) than subscribed customers (33.49). Study the newsletter's value proposition and test targeted incentives to promote subscription and improve customer engagement.

- **Evaluate discount strategies:** Hats, sneakers, coats, pants, and sweater had discount rates ranging from approximately 46% to 50%. Assess the profitability of these products and determine whether discounts are necessary at their current frequency.

- **Strengthen customer retention:** Utilize purchase history to identify returning and loyal customers and develop targeted retention initiatives, such as personalised promotions and loyalty rewards. Monitor repeat purchasing and subscription rates to evaluate their effectiveness.

7. REFERENCES AND ACKNOWLEDGEMENTS

This project was undertaken as a learning exercise, using the dataset and project resources provided by Amlan Mohanty in his GitHub repository and accompanying YouTube tutorial. These resources served as a reference for understanding the retail customer shopping analysis workflow.

The dataset was independently cleaned and modified using Python in Jupyter Notebook, analysed using SQL in MySQL, and visualised using Power BI. The SQL queries, dashboard, and report were developed as part of my own implementation of the project.

References

- Mohanty, A. (n.d.). *Customer Trends Data Analysis: SQL, Python and Power BI* [GitHub repository]. GitHub. https://github.com/amlanmohanty1/customer-trends-data-analysis-SQL-Python-PowerBI
- Mohanty, A. (n.d.). *Customer Shopping Trends Data Analysis Project* [YouTube video]. YouTube. https://www.youtube.com/watch?v=5PrZvPeUw60